

Editorial review: Modeling mRNA populations (PLOS Genetics draft)

Internal pre-submission review, prepared with Claude Code for the coauthors

2026-10-09

Scope of this review

This review treats the manuscript as a PLOS Genetics editor would at initial submission. It covers the current LaTeX source (`mRNA_Populations_PLOS_GENETICS.tex`, last edited 2025-08-02), the supplemental text, the figures, and the reference list. The committed PDF was built on 2024-08-12 from an older draft and does not reflect the current source, so all comments below refer to the source and to a fresh compile of it.

Version 1.1 (2026-10-09). Adds Section 9, a check of the R code in Ricardo’s **Ribosome** repository (commit f70c00a, 2023-07-25), and updates items 1.5, 3.2, 3.3 and 3.4, which that check resolves. Version 1.0 was written from the manuscript alone. **Version 1.2 (2026-10-09).** Corrects 1.5 and 9.7: the repository is public, not private.

Decision

Return to authors before review. The model is clean and the central idea (two coupled polysome populations, capped and decapped) is worth publishing. In its current form, however, the manuscript is incomplete, the single validation is weak, the prior-art framing is incorrect, and there are internal inconsistencies between the equations and the stated results. An editor would most likely triage it for insufficient strength of advance rather than send it out.

The code check (Section 9) sharpens this. The headline result, that a short-lived mRNA produces just under half its protein from decapped transcripts, is computed at an elongation rate of one codon per second and at a decapping rate three times smaller than the figure caption states. At the captioned rate the model gives two thirds, and at realistic yeast elongation rates it gives one sixth to one third. The central claim therefore has to be recomputed before the paper goes anywhere.

Two paths forward:

1. **PLOS Genetics.** Add a genetics-facing result (see Section 4) and a second independent validation. Cope, Pak and Gilchrist (2026, PLOS Genetics) shows the journal accepts ribosome-scale modeling with a genomic application, so the fit is possible.
2. **PLOS Computational Biology.** The more natural home for a theory paper with one correlational validation. Chevalier et al. (2023), the closest prior model, is published there.

1. Completeness and build state

1.1. **The committed PDF is stale.** Built 2024-08-12 with a different abstract and introduction. Rebuild before anyone reads it.

1.2. **Six citation keys have no bibliography entry** and render as [?]: MacDonaldEtAl1968, MacDonaldAndGibbs1969, vonHeijneEtAl1987, Chou2003, GilchristAndWagner2006, ShahAndGilchrist2011.

1.3. **Compile errors in the current source.** Label `eqn:S` is referenced but the label is `eq:S`. `\mvechatstart` is an undefined macro (should be `\mvechatstar`). Two uses of `\MRL^*` in text mode produce “Missing \$” and a 1098 pt overfull line in the Results section on protein production. The `strip` environment errors in one-column mode.

1.4. **Placeholders visible in the body.** Author Summary reads “NEED TO WRITE THIS”. A bold “[Mike: Ricardo can you revise ...]” note and an `\mmpar` margin note are still in the Introduction.

1.5. **Code repository is incomplete.** `github.com/rurquidi/Ribosome` is public (v1.0 and the first draft of v1.1 called it private; it returned 404 at the time of checking and was public by the end of the day). Submission needs a Data Availability Statement pointing at it. The repository lacks the input data: the scripts read Presnyak 2015, Chan 2018, Weinberg 2016 RPKMs, Dao Duc and Song 2018 rates, and the yeast and Arabidopsis FASTA files from `../Data` and `../../Data`, which were never pushed. The package depends on `limSolve`, has a placeholder DESCRIPTION (no license, template description text), and a top-level `polysome_model_results.Rmd` that is the unedited RStudio template. See Section 9.

1.6. **Bibliography entries never cited:** RN1 (Browning and Bailey-Serres 2015) and RN37 (Wu and Jaffrey 2016).

2. PLOS Genetics format requirements

2.1. Materials and Methods must follow the Discussion unless the cover letter justifies otherwise. The draft places Methods before Results.

2.2. Author Summary: 150 to 200 words, first person, written for a non-specialist. Missing.

2.3. Short title (70 characters) is missing. The full title “Modeling mRNA populations” is too generic to be findable. Say what is coupled and what is predicted.

2.4. References must be numbered in order of first citation, Vancouver style, with full DOIs. The hand-written `thebibliography` is alphabetical and has no DOIs.

2.5. Initial submission should be double spaced with continuous line numbers. The abstract and summary currently sit outside `\linenumbers`.

2.6. Supplemental text errors. Matrix U: the second diagonal entry lacks the leading minus sign and parentheses. Matrix R: stray backtick in the (0,1) entry; the last two diagonal entries use (imax-2)/imax where (imax-1)/imax is meant.

3. Model correctness (verify each against the R code)

3.1. **ODE typos.** `\kappa0` in the general capped row. That row’s coefficient reads `kappa_0 (i-1)/imax` but should be `kappa_0 (1 - (i-1)/imax)`. The last decapped row swaps `m*_imax` and `m_imax`.

3.2. **Eq. (decapped_abundance) appears to drop a factor of imax.** Summing the decapped equations from class `k` upward gives

$$\text{tau } (k/\text{imax}) \text{ m}^*_k = \mu * \text{sum}_{\{j \geq k\}} \text{m}_j$$

so $m^*_k = (\text{imax}/k)(\mu/\text{tau}) S_k$, not $(1/k)(\mu/\text{tau}) S_k$. The total decapped pool, the odds expression, and p^*_i all inherit this. *Resolved in v1.1*: the code divides by $\text{tau}_i = \text{tau}_c i/(9 \text{ imax})$, so it carries the `imax` factor. The text is wrong and the figures are right. Insert `imax` in Eq. (decapped_abundance), Eq. (decapped_solution), the total decapped pool, Eq. (odds), and Eq. (decapped_distribution).

3.3. **An exact result the paper misses.** Multiply the identity in 3.2 by $\text{tau } k/\text{imax}$ and sum over `k`:

$$\begin{aligned} \text{decapped protein production} &= \mu * \text{sum}_j j \text{ m}_j \\ \text{capped protein production} &= (\text{tau}/\text{imax}) * \text{sum}_j j \text{ m}_j \end{aligned}$$

So the decapped share of total protein output is

$$(\mu \text{ imax} / \text{tau}) / (1 + \mu \text{ imax} / \text{tau})$$

a one-parameter closed form, independent of κ . It is the probability that decapping happens before a ribosome finishes its transit. This is a cleaner headline than the “41 percent” figure, and it shows the claim “never more than half” is a statement about the parameter range, not a theorem: the share exceeds one half whenever the capped half-life is shorter than $\ln(2)$ times the ribosome transit time imax/tau . *Confirmed numerically in v1.1* (Section 9.2): a re-implementation of the coded model gives a share that is constant in κ to four decimal places and matches the formula.

3.4. **Fig 10 panels B and C are swapped, and the mu values in the caption and legend are wrong.** *Confirmed in v1.1 from the figure code* (Section 9.3). The three curves are computed at $\mu = 2.2\text{e-}4$, $3.3\text{e-}4$ and $2.3\text{e-}3$ per second (half-lives 52, 35 and 5 minutes), not at $2.2\text{e-}4$, $1.7\text{e-}3$ and $5.7\text{e-}3$ (52, 7 and 2 minutes) as captioned. The “41 percent” in the Results and Discussion is the share at $\mu = 2.3\text{e-}3$.

3.5. **The supplement’s lambda/mu result is presented as discovered by “manual exploration”.** It follows in one line from summing the capped ODEs: $d/dt (\text{sum } m_i) = \text{lambda} - \mu \text{ sum } m_i$. State it that way.

3.6. **Units.** The initiation-to-elongation ratio κ' is dimensionless, not in units of 1/s. Its range is given as 0.001 to 0.1 in Methods, but the figures span $1\text{e-}4$ to 0.5.

3.7. **The delta assumption is stated two ways:** $\text{delta} \gg \text{tau}$ in Methods, $\text{delta} \gg \text{lambda}$ in Discussion. Pick one and justify it.

3.8. **Yeast decapping rate statistics are inconsistent.** Methods: mean $1.3\text{e-}3$ (SD $1.8\text{e-}3$). Figures: median $1.7\text{e-}3$. A right-skewed rate distribution should have mean above median. The Arabidopsis range “ $1.7 \times (10^{-4} \pm 2 \times 10^{-4})$ ” is garbled.

3.9. **Percentile language is inconsistent.** $2e-4/s$ is called the “99th percentile” (of half-life) in Results; $5.7e-3$ is called “low decapping” once in the Discussion. Use one convention, percentiles of μ .

4. Strength of advance and validation

4.1. **The only test against data is Fig 11**, and the text overstates it (“coincide”). Spearman $\rho = 0.59$ is shown in the figure but not stated in the text. Observed loads sit well above predicted ones:

Quantity	Predicted	Observed
Mean ribosome load range	1 to 11	1.6 to 40

The absolute scale of the empirical load rests on an ad hoc “200 / length” correction. Present this as a rank test, explain the roughly three-fold offset, and add a second independent check. Arava et al. (2003) polysome-profile data are the obvious one; they also report that ribosome density falls with ORF length, a direct test of the imax prediction.

4.2. **The empirical μ distribution drives the headline result, and it is method dependent.** Chan et al. (2018) report non-invasive yeast half-lives averaging 4.8 minutes, far shorter than Presnyak et al. (2015), and conclude that initiation and decay compete. Shorter half-lives shift μ upward and make the decapped share larger. Discuss this.

4.3. **Say what a geneticist does with the model.** For example, that measured half-lives and polysome profiles systematically mix two populations, and how the model separates them.

5. Ribosome drop-off and nonsense errors (to add)

5.1. **The model assumes every initiated ribosome completes the coding sequence**, in both the capped and decapped states. This assumption is implicit; it is never stated. Premature termination, drop-off, and nonsense errors are not mentioned anywhere in the text, the parameter table, or the Limitations. NMD and no-go decay appear only as excluded endonucleolytic pathways.

5.2. **Citation.** Cope AL, Pak D, Gilchrist MA. The importance of nonsense errors: Estimating the rates and implications of ribosome drop-off during protein synthesis. PLoS Genet. 2026;22(6):e1012162. doi:10.1371/journal.pgen.1012162. Preprint: bioRxiv 2024.09.05.611510. Drop-off was already present in the per-transcript models of Gilchrist (2005, J Theor Biol) and Gilchrist and Wagner (2006), which the draft cites but has not entered in the bibliography.

5.3. **Where to add it**, in increasing effort.

- Introduction, TASEP paragraph: note that drop-off is part of the earlier per-transcript models and that Cope et al. provide current per-codon estimates from yeast ribosome profiling.
- Methods: state the completion assumption explicitly. Limitations: say drop-off is omitted, why, and its magnitude per Cope et al.
- Model extension: drop-off adds a transition from class i to class $i-1$ at rate i times the per-ribosome drop-off rate, in both states, returning a ribosome without producing protein. The system stays linear and tridiagonal, so `solve.tridiag` still applies. It modifies the exact

result in 3.3, because ribosomes on decapped mRNAs spend longer in transit and are more exposed to drop-off.

5.4. Expected size of the effect. At roughly one error per ten thousand codons and a median $i_{\max}$ of 39 (about 350 codons), the loss per transit is a few percent. A short paragraph giving the first-order correction would likely suffice instead of a full re-analysis.

5.5. Unit conversion. Cope et al. estimate drop-off per codon; this model’s length unit is one ribosome footprint of nine codons. The conversion is the same one the paper already uses for τ and should be stated in the same sentence.

6. References: framing and missing prior art

6.1. “Few have explored the interaction” is not defensible. Directly relevant models not cited:

- Nagar A, Valleriani A, Lipowsky R. Translation by ribosomes with mRNA degradation: exclusion processes on aging tracks. *J Stat Phys.* 2011;145:1385-1404.
- Deneke C, Lipowsky R, Valleriani A. Effect of ribosome shielding on mRNA stability. *Phys Biol.* 2013;10:046008.
- Chevalier C, et al. Physical modeling of ribosomes along messenger RNA: estimating kinetic parameters from ribosome profiling experiments using a ballistic model. *PLoS Comput Biol.* 2023;19(10):e1011522. Includes degradation and predicts monosome and polysome distributions.
- Edri S, Tuller T. Quantifying the effect of ribosomal density on mRNA stability. *PLoS ONE.* 2014;9(7):e102308.
- Dave P, et al. Single-molecule imaging reveals translation-dependent destabilization of mRNAs. *Mol Cell.* 2023;83(4):589-606. Contains a model of ribosome-flux-dependent decay.

6.2. Empirical coupling of translation and decay, 2018 to 2025, is absent.

- Chan LY, et al. Non-invasive measurement of mRNA decay reveals translation initiation as the major determinant of mRNA stability. *eLife.* 2018;7:e32536.
- Bicknell AA, et al. Attenuating ribosome load improves protein output from mRNA by limiting translation-dependent mRNA decay. *Cell Rep.* 2024;43(4):114098. This is the applied point the Discussion gestures at.
- Koegel A, et al. Structural basis of mRNA decay by the human exosome-ribosome supercomplex. *Nature.* 2024;635:237-242. For the 3’ pathway the model excludes.
- Collart MA, Audebert L, Bushell M. Roles of the CCR4-Not complex in translation and dynamics of co-translation events. *WIREs RNA.* 2023;15(1):e1827.
- Hanson G, Collier J. Codon optimality, bias and usage in translation and mRNA decay. *Nat Rev Mol Cell Biol.* 2018;19:20-30.

6.3. Plant co-translational decay literature is thin for a paper that uses Arabidopsis parameters.

- Yu X, Willmann MR, Anderson SJ, Gregory BD. Genome-wide mapping of uncapped and cleaved transcripts reveals a role for the nuclear mRNA cap-binding complex in cotranslational RNA decay in Arabidopsis. *Plant Cell.* 2016;28(10):2385-2397.
- Carpentier MC, et al. 5’ to 3’ cotranslational mRNA decay is dynamically regulated during Arabidopsis seedling development and fine-tunes translation efficiency. *Plant Physiol.*

2020;184(3):1251-1262.

- Deragon JM, Merret R. Co-translational mRNA decay in plants: recent advances and future directions. *J Exp Bot.* 2025. doi:10.1093/jxb/eraf146.
- Wu HL, et al. Improved super-resolution ribosome profiling reveals prevalent translation of upstream ORFs and small ORFs in Arabidopsis. *Plant Cell.* 2024;36(3):510-539.

6.4. **Parameter sources for kappa and tau.** Beside Dao Duc and Song (2018), cite Riba A, et al. Protein synthesis rates and ribosome occupancies reveal determinants of translation elongation rates. *PNAS.* 2019;116(30):15023-15032, and Erdmann-Pham DD, Dao Duc K, Song YS. The key parameters that govern translation efficiency. *Cell Syst.* 2020;10(2):183-192.

6.5. **Odd choices.** RN18 and RN19 (Wu and Tian, generic multistep stochastic models) and RN20 (selenoprotein knockdown) are cited as “models of mRNA degradation”. Replace with the Valleriani line above and Hanson and Collier (2018). Ensembl release 109 and R 3.6 should be updated; data.table is cited at a 2021 version.

7. Figures

7.1. Fig 11 prints rho to seven digits. Use two significant figures and state n and the test in the caption.

7.2. Fig 10 legend gives half-lives in minutes while the caption gives mu in 1/s. Use one. The legend text “(quantile))” has an extra parenthesis.

7.3. Fig 2 (parameter histograms) could move to the supplement. Figs 3 and 4 share one layout and would read better as a single figure.

8. Writing

8.1. A bad find-and-replace has damaged words: “Th such”, “Thwever”, “Th m*_0”, “Cois”, “Coft” (for “As such”, “However”, “If”, “This”, “Left”). Search the source for these.

8.2. The abstract’s first sentence has two grammatical errors. “Novel” appears four times. “Impressive” (Results) is not a result.

8.3. The second Introduction paragraph on NMD and no-go decay is flagged in a margin note as out of place. Agreed; fold it into Limitations and pair it with the drop-off discussion in Section 5.

9. Code verification (added in v1.1)

The R code lives in the repository `rurquidi/Ribosome` (public; 11 commits, 2020-05-10 to 2023-07-25). The solver is `R/Polysome_functions_RAUC.R`; the paper’s figures are produced by `Text/Paper final figures.Rmd` (1700 lines), not by the scripts under `R/`. `R/Manuscript_Figures.R` (3954 lines) is an earlier, partly superseded version of the same material. I re-implemented the coded model in base R (40 by 40 linear solve, no `limSolve`) and reproduced the Fig 10 curves to three figures.

9.1. **The code matches the ODEs in the manuscript.** Initiation is $\kappa(1 - i/i_{\max})$, termination from class i is $\tau_c i / (9 i_{\max})$ with τ_c in codons per second, and the decapped classes are $m^*_i = \mu S_i / \tau_i$, with $m^*_0 = \mu S_0 / \delta$. This confirms 3.2.

9.2. **The decapped share of protein output is exactly $9 \mu \text{imax}/\tau_c$ over $1 + 9 \mu \text{imax}/\tau_c$,** independent of kappa. The re-implementation gives the same share at every kappa from $1e-4$ to 0.5 . This confirms 3.3. The paper should state the closed form and drop the “never more than half” claim, which fails at the paper’s own high-decapping value (see 9.4).

9.3. **Fig 10 panel assignment.** The parameter grid is built with `expand.grid`, so kappa varies fastest and each block of 5000 rows is one μ value. The figure code plots these blocks:

Panel as drawn	Rows plotted	Actual μ (1/s)	Half-life	Caption	Peak, code	Peak, figure
A top curve, panel C	45001-50000	$2.2e-4$	52 min	$2.2e-4$, 52 min	1.00	1.0
A middle curve, panel B	40001-45000	$3.3e-4$	35 min	$1.7e-3$, 7 min	0.69	0.69
A bottom curve, panel D	20001-25000	$2.3e-3$	5 min	$5.7e-3$, 2 min	0.14	0.14

The low-decapping plot (peak 1.0) is panel C in the composite and the mid-decapping plot (peak 0.69) is panel B, so the panels were swapped when the figure was assembled. The decapped share at $\mu = 2.3e-3$ is 45 percent, which is the “41 percent” quoted in the text.

9.4. **Every model figure assumes an elongation rate of one codon per second.** All figure runs set $\tau = 1$ and treat kappa as kappa prime, but μ is entered in absolute units and is never scaled by τ . So μ/τ_c is a second dimensionless group that the paper’s “collapse to one parameter” argument (Methods, Data Sources) does not acknowledge, and the figures implicitly make yeast decapping 5 to 10 times faster relative to elongation than it is. The effect on the headline:

τ_c (codons/s)	Decapped share, $\mu = 5.7e-3$, $\text{imax} = 39$
1 (as coded)	67 percent
5	29 percent
10	17 percent

At the captioned μ the share exceeds one half, contradicting the Discussion. At realistic yeast elongation rates (Dao Duc and Song 2018, Riba et al. 2019) it is a sixth to a third. The validation in Fig 11, by contrast, uses gene-specific τ_c and kappa from Dao Duc and Song, so the model figures and the validation are not on the same footing. Fix: present results in terms of the two dimensionless groups kappa/τ_c and $9 \mu \text{imax}/\tau_c$, or rerun the figures with a realistic τ_c and μ scaled by it.

9.5. **Fig 11 is coded as described.** Empirical load is RPF RPKM over mRNA RPKM times length over 200, with length taken as 27 imax nucleotides; model load uses per-gene kappa and τ_c from Dao Duc and Song and μ from Presnyak; the test is Spearman on log10 values. The reporting issues in 4.1 and 7.1 stand.

9.6. **Protein production in the code is total bound ribosomes** (sum of $i m_i$), which is proportional to the production rate only at fixed i_{max} . That is correct for Fig 10 but would be wrong if reused across the i_{max} range.

9.7. **Reproducibility.** The input data directory was not pushed (see 1.5), `limSolve` is required, the solver exists in three near-duplicate variants (`CalcMarkedClass`, `_newtau`, `NP`) of which the figure code calls the first, `CalcMarkedClass` recomputes the capped solution with `CalcUnmarkedClass` rather than the `_newtau` variant when none is passed, and the figure code selects parameter sets by hard-coded row ranges. None of this blocks publication, but the repository needs a README that maps each figure to its chunk and parameter block, a license, and the data files before it is cited in a submission.

Sources consulted

PLOS Genetics submission guidelines and journal information pages (journals.plos.org/plosgenetics), PubMed, bioRxiv, the publisher pages for each reference listed above, and the `rurquidi/Ribosome` repository at commit `f70c00a` (cloned to `~/Repositories/Ribosome` on 2026-10-09).